Tristan Bilot

Paris, FR | Email | Website | LinkedIn | GitHub | Google Scholar

Education

Ph.D. in Computer Science, Université Paris-Saclay, Paris, France — 2022 – 2025
- Thesis: Detecting Advanced Cyberattacks with Self-Supervised Graph Learning
- Supervisors: Pr Khaldoun Al Agha, Dr Nour El Madhoun

Master's (Diplôme d'Ingénieur) in Computer Science, EPITA, Paris, France — 2019 – 2022
- GPA: 4.0/4.0
- Focus on Computer Security, Systems, Network

DUT in Computer Science, Université Paris-Cité, Paris, France — 2017 – 2019
- Focus on Algorithms, Data Structures, Data Mining, Reflective Programming

Experience

Visiting Scholar – University of Oxford, Oxford, UK — Sept. 2026 –
- Conducting research on intrusion detection at Oxford with Thomas Pasquier

Postdoctoral Fellow – University of British Columbia, Vancouver, BC, Canada — Feb. 2026 –
- Research on foundation models for security applications
- Funded by an Amazon Research Award

Applied Scientist Intern – Amazon Web Services (AWS), New York, NY, USA — Oct. 2025 – Jan. 2026
- Part of the Security Analytics and Artificial Intelligence Research team (SAAR)
- First-authored a NeurIPS paper on multi-agent system security
- Declined two full-time offers in the US

Ph.D. Researcher – Iriguard, LISN, LISITE, Paris, France — Oct. 2022 – Oct. 2025
- Developed scalable intrusion detection systems with deep learning on client data, focus on APTs and zero-days
- Ph.D. funded by Iriguard and in collaboration with LISN and LISITE labs

Visiting Research Student – University of British Columbia, Vancouver, BC, Canada — Apr. – Jun. 2024
- Research in provenance-based intrusion detection systems with GNNs and self-supervised learning, supervised by Thomas Pasquier
- Worked on large-scale temporal graphs and robustness to adversarial attacks

Student Researcher – EPITA Systems Laboratory (LSE), Paris, France — Sep. 2021 – Aug. 2022
- Research on GNNs for phishing web page detection, supervised by Dr. Badis Hammi

Data Engineer Apprentice – Carrefour-Google AI Lab, Paris, France — May 2021 – Aug. 2022
- Deployed ML models in production + optimized training time (4x and 3x faster)
- Built a scalable BigQuery fetching tool, presented in internal engineering reviews
- Deployed a multi-project data pipeline with Airflow, dbt, GCP, Kubernetes

Software Engineer Apprentice – Carrefour, Paris/Massy, France — Sep. 2019 – Apr. 2021
- Developed new features for the Carrefour iOS app (1.5M+ monthly users)
- Integrated Apple Wallet into the app

Software Engineer Intern – Micropole, Levallois-Perret, France — May 2019 – Aug. 2019
- Developed backend features for websites and web services
- Improved website loading speed by ∼30%

Publications

Full list: Google Scholar – *AR: Acceptance Rate*

Learning Transferable Representations from Operating System Entities via Provenance Graph Distillation — *NeurIPS 2026 (AR: 25.7%)* [paper, code]
Tristan Bilot, Xueyuan Han, Thomas Pasquier

PURPLEAUDIT: A Co-Evolutionary Auditing Framework for Task Hijacking in Multi-Agent Systems — *NeurIPS 2026 (AR: 25.7%)* [paper]
Tristan Bilot, Zhilu Zhang, Kay Liu, Mikhail Kuznetsov, Wei Din

LARES: Targeted Lateral Movement Detection in Evolving Networks Through Source Host Identification — *ACSAC 2026 (AR: 19%)* [paper, code]
Tristan Bilot, Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui, Thomas Pasquier

PIDSMaker: Building and Evaluating Provenance-based Intrusion Detection Systems — *KDD 2026 (AR: 21%* [paper, code, poster]
Tristan Bilot, Baoxiang Jiang, Thomas Pasquier

Sometimes Simpler is Better: A Comprehensive Analysis of State-of-the-Art Provenance-Based Intrusion Detection Systems — *USENIX Security 2025 (AR: 17.1%)* [paper, code, poster, slides, video]
Tristan Bilot, Baoxiang Jiang, Zefeng Li, Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui, Thomas Pasquier

ORTHRUS: Achieving High Quality of Attribution in Provenance-based Intrusion Detection Systems — *USENIX Security 2025 (AR: 17.1%)* [paper, code, slides, video]
Baoxiang Jiang, Tristan Bilot, Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui, Shahrear Iqbal, Xueyuan Han, Thomas Pasquier

Few Edges Are Enough: Few-Shot Network Attack Detection with Graph Neural Networks — *IWSEC 2024 (AR: 29.8%),* ***best paper*** [paper, code, slides]
Tristan Bilot, Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui

A Survey on Malware Detection with Graph Representation Learning — *ACM Computing Surveys, 2024* [paper]
Tristan Bilot, Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui

A Benchmark of Graph Augmentations for Contrastive Learning-Based Network Attack Detection with Graph Neural Networks — *CSNet 2023* [paper, poster]
Tristan Bilot, Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui

Graph Neural Networks for Intrusion Detection: A Survey — *IEEE Access, 2023* [paper]
Tristan Bilot, Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui

PhishGNN: A Phishing Website Detection Framework using Graph Neural Networks — *SECRYPT 2022* [paper, code, slides]
Tristan Bilot, Grégoire Geis, Badis Hammi

Publications (Under review)

PERCEPTOR: Temporal Graph Transformer for Large-Margin Provenance-based Intrusion Detection — *Under review, NDSS 2026*

Prizes and Awards

Impact Award, Starthese Start'up Pitch-iD – national competition in which doctoral researchers pitch their research as a deep-tech venture; carried an invitation to Bpifrance's BIG event and to a learning expedition at MIT — 07/07/2026

Best Paper Award, IWSEC 2024 – *Few Edges Are Enough* — 17/09/2024

Technical Articles

USENIX ;login: – Article based on our two USENIX Security 2025 papers — 2025 [article]

Medium – Various articles on MLX/CUDA benchmarks, data eng., software eng. — 2020–now [11 articles + code]

Personal Blog – "Deep learning from scratch" series, on autodiff & backpropagation — 2022 [5 articles, code]

Invited Talks

BIG 2026 – Pitch at the largest European's entrepreneurship event by Bpifrance — France – 2026 [event]

Journées Innovation Évry 2026 – Deeptech entrepreneurship pitch — France – 2026 [event]

Ecole Polytechnique de Louvain – PIDSMaker Tutorial — Belgium – 2026 [slides]

NDSS'26 PRISM Workshop – PIDSMaker Tutorial — USA – 2026 [slides, video]

Télécom Paris – Class on AI techniques for advanced attack detection — France – 2026 [slides]

The University of Texas at El Paso – Achieving High Quality of Attribution in IDS — USA – 2025 [slides]

GenAI Meetup Morocco – How AI protects us from cyberattacks? — Morocco – 2025 [slides]

EPITA Seminar – Introduction to ORTHRUS and PIDSMAKER — France – 2025 [slides]

University of British Columbia – Inductive Host Detection in Large Temp. Graphs — Canada – 2024 [slides]

Institut Mines-Télécom – System-level IDS with Graph Neural Networks — France – 2024 [slides]

DATAIA Day Saclay – Detecting Complex Attacks with Graph Deep Learning — France – 2022 [poster]

Carrefour – Data Engineering Applied to Retail — France – 2022 [slides]

Projects

PIDSMaker – Deep learning framework for building provenance-based IDS — 2025– [code]

Apple MLX – Implemented backpropagation of scattering operations in C++ — 2024– [code]

MLX-graphs – GNN library on top of MLX with optimized GPU kernels — 2024– [code]

MLX-benchmark – Benchmark framework for MLX, Apple chips and CUDA GPUs — 2024– [code]

Deepiler – Transformer-based decompiler to convert binaries into C code — 2022 [code]

K – x86 Kernel – Simple kernel written in C and Assembly x86 — 2021 [code]

Program Committees

IEEE Symposium on Security and Privacy (IEEE S&P) — 2027

Learning on Graphs (LoG) — 2026

IEEE Transactions on Networking — 2025

KES Detection of Complex Attacks — 2025

Information Security Journal: A Global Perspective — 2025

Skills

Programming: Python, Swift, C, C++, Bash, JS, Java, CUDA, Assembly x86, Rust
ML Frameworks: PyTorch, MLX, Jax/Haiku, pandas, scikit-learn
Agentic AI: Autogen/AG2, AWS Bedrock, LangChain, LangGraph
ML Skills: Self-supervised learning, GNNs, Transformers, LLM fine-tuning (LLaMA+QLoRA), GPU parallelization & vectorization, framework coding, large-scale training under limited resources
Infrastructure: GCP, AWS, Docker, Kubernetes, Airflow, dbt, W&B
Languages: French (native), English (fluent), Spanish (notions)

Activities & Interests

Volunteering: Protection civile (2018–2022), first aid
Academic: Student bureau, Junior Entreprise, class representative, hackathons (Google HashCode, Design4Green, Carrefour)
Hobbies: Music (DJ mix), cosmology, traveling, surfing